

HDOC – Experiment 13

Question

Develop a menu-driven C program that accepts a positive integer and performs one of the following operations based on the user's choice:

- **Count the number of even and odd digits.**
- **Find the sum of even digits and the sum of odd digits separately.**

❖ Understanding the problem

Input

- Positive Integer
- User's Menu Choice

Output

Option 1

- Number of Even Digits
- Number of Odd Digits

Option 2

- Sum of Even Digits
- Sum of Odd Digits

❖ Test case table

Test Case	Input (Choice, Number)	Expected Output	Actual Output	Result
1	1, 12045	Even Digits = 3, Odd Digits = 2	Same	Pass
2	2, 12345	Sum Even = 6, Sum Odd = 9	Same	Pass
3	1, 24680	Even Digits = 5, Odd Digits = 0	Same	Pass
4	2, 13579	Sum Even = 0, Sum Odd = 25	Same	Pass
5	1, 987654	Even Digits = 3, Odd Digits = 3	Same	Pass

❖ Algorithm

1. Start.
2. Display the menu with two options.
3. Read the user's choice.
4. Read the positive integer.
5. Copy the number into a temporary variable.
6. Initialize counters and sums to zero.
7. According to the menu choice:
 - 7.1 **Case 1:** Traverse each digit and count even and odd digits.
 - 7.2 **Case 2:** Traverse each digit and add even digits to the even sum and odd digits to the odd sum.
8. Display the required result.
9. If the menu choice is invalid, display an error message.
10. Stop.

❖ Program in C

```
#include <stdio.h>

int main()
{
    int choice;

    int number, temp, digit;

    int evenCount = 0, oddCount = 0;
    int evenSum = 0, oddSum = 0;

    printf("==== MENU =====\n");
    printf("1. Count Even and Odd Digits\n");
    printf("2. Find Sum of Even and Odd Digits\n");

    printf("Enter your choice: ");
    scanf("%d", &choice);

    printf("Enter a positive integer: ");
    scanf("%d", &number);

    temp = number;

    switch(choice)
    {
        case 1:

            while(temp > 0)
            {
```

```
    digit = temp % 10;

    if(digit % 2 == 0)
        evenCount++;
    else
        oddCount++;

    temp = temp / 10;
}

printf("\nNumber of Even Digits = %d\n", evenCount);
printf("Number of Odd Digits = %d\n", oddCount);

break;

case 2:

    while(temp > 0)
    {
        digit = temp % 10;

        if(digit % 2 == 0)
            evenSum += digit;
        else
            oddSum += digit;

        temp = temp / 10;
    }
```

```

printf("\nSum of Even Digits = %d\n", evenSum);
printf("Sum of Odd Digits = %d\n", oddSum);

break;

default:

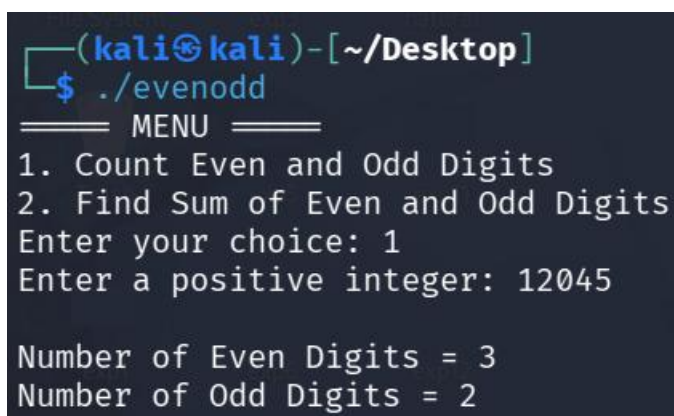
printf("\nInvalid Choice!\n");
}

return 0;
}

```

❖ Compile, Execute and Evaluate

Test case 1



```

(kali@kali)-[~/Desktop]
$ ./evenodd
===== MENU =====
1. Count Even and Odd Digits
2. Find Sum of Even and Odd Digits
Enter your choice: 1
Enter a positive integer: 12045

Number of Even Digits = 3
Number of Odd Digits = 2

```

Test case 2

```
(kali㉿kali)-[~/Desktop]
$ ./evenodd
===== MENU =====
1. Count Even and Odd Digits
2. Find Sum of Even and Odd Digits
Enter your choice: 2
Enter a positive integer: 12345

Sum of Even Digits = 6
Sum of Odd Digits = 9
```

Test case 3

```
(kali㉿kali)-[~/Desktop]
$ ./evenodd
===== MENU =====
1. Count Even and Odd Digits
2. Find Sum of Even and Odd Digits
Enter your choice: 1
Enter a positive integer: 24680

Number of Even Digits = 5
Number of Odd Digits = 0
```

Test case 4

```
(kali㉿kali)-[~/Desktop]
$ ./evenodd
===== MENU =====
1. Count Even and Odd Digits
2. Find Sum of Even and Odd Digits
Enter your choice: 2
Enter a positive integer: 13579

Sum of Even Digits = 0
Sum of Odd Digits = 25
```

Test case 5

```
(kali㉿kali)-[~/Desktop]
$ ./evenodd
===== MENU =====
1. Count Even and Odd Digits
2. Find Sum of Even and Odd Digits
Enter your choice: 1
Enter a positive integer: 987654
code2 swap_menu.c
Number of Even Digits = 3
Number of Odd Digits = 3
```

❖ Compilation and execution screenshot

```
(kali㉿kali)-[~/Desktop]
$ vi evenodd.c

(kali㉿kali)-[~/Desktop]
$ gcc evenodd.c -o evenodd

(kali㉿kali)-[~/Desktop]
$ ./evenodd
===== MENU =====
1. Count Even and Odd Digits
2. Find Sum of Even and Odd Digits
Enter your choice: 1
Enter a positive integer: 12045
Number of Even Digits = 3
Number of Odd Digits = 2

(kali㉿kali)-[~/Desktop]
$ ./evenodd
===== MENU =====
1. Count Even and Odd Digits
2. Find Sum of Even and Odd Digits
Enter your choice: 2
Enter a positive integer: 12345
Sum of Even Digits = 6
Sum of Odd Digits = 9

(kali㉿kali)-[~/Desktop]
$ ./evenodd
===== MENU =====
1. Count Even and Odd Digits
2. Find Sum of Even and Odd Digits
Enter your choice: 1
Enter a positive integer: 24680
Number of Even Digits = 5
Number of Odd Digits = 0
```

```
(kali㉿kali)-[~/Desktop]
$ ./evenodd
===== MENU =====
1. Count Even and Odd Digits
2. Find Sum of Even and Odd Digits
Enter your choice: 2
Enter a positive integer: 13579

Sum of Even Digits = 0
Sum of Odd Digits = 25
```

```
(kali㉿kali)-[~/Desktop]
$ ./evenodd
===== MENU =====
1. Count Even and Odd Digits
2. Find Sum of Even and Odd Digits
Enter your choice: 1
Enter a positive integer: 987654

Number of Even Digits = 3
Number of Odd Digits = 3
```